
An Analysis of Trends and Patterns of Major Diseases and Self-Harm in India and Globally

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Background

Non-communicable diseases (NCDs) such as cancer, stroke, diabetes, and hypertension, coupled with critical mental health issues leading to self-harm, represent a significant and growing public health challenge in India and globally. Understanding the trends, demographic risk factors, and geographical distribution of these health issues is essential for developing effective public health policies.

This project aims to conduct a comprehensive data-driven analysis of these critical health indicators.

The analysis seeks to answer the following key questions:

- What are the leading causes of deaths in each State and Union Territory of India?
 - What have been the historical trends in cancer mortality rates and absolute death counts in India since 1990?
 - Which specific types of cancer (e.g., stomach, breast, leukaemia) are the leading causes of cancer-related mortality in India?
 - Based on historical data, what are the projected numbers of cancer deaths in India for the upcoming decade?
 - What are the recent global trends in stroke incidence and diabetes prevalence?
 - Which age groups are most affected by diabetes?
 - How has the number of deaths due to self-harm in India evolved in recent years?
 - Is there a significant difference in the number of deaths from self-harm between males and females in India?
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Data Collection Methodology

The data has collected from

- Scrapped from StatisticsTimes: <https://statisticstimes.com/demographics/country/india-population.php>
 - Datasets provided by Institute of Health Metrics and Evaluation : <https://www.healthdata.org/research-analysis/gbd>
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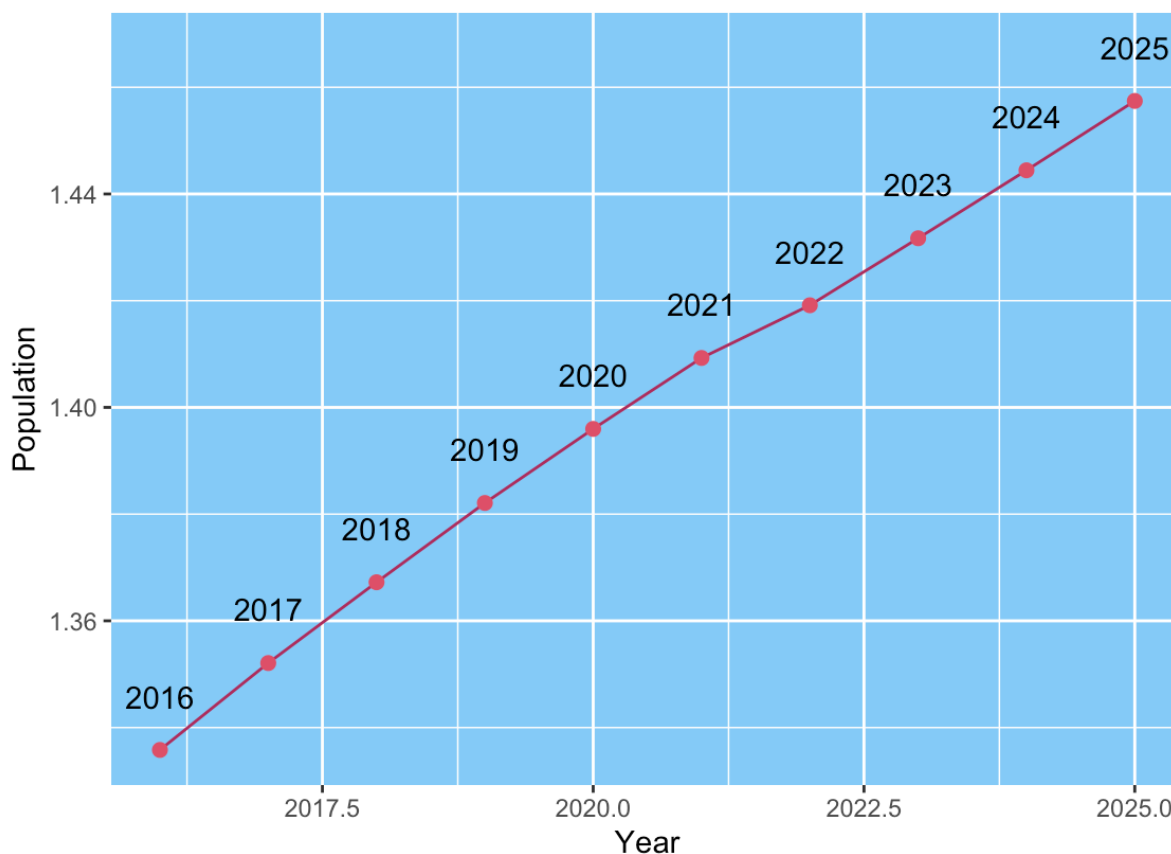
Ethics Check

All data has been ethically collected. StatisticsTimes allows web scrapping. No overloading of servers is done. Only necessary data has been scrapped. No copyrighted data has been extracted.

IHME an independent global health research centre at the University of Washington (Seattle, USA). IHME collects, analyses, and shares data on health problems around the world. It allows download of datasets for personal or educational use. No personal details are collected.

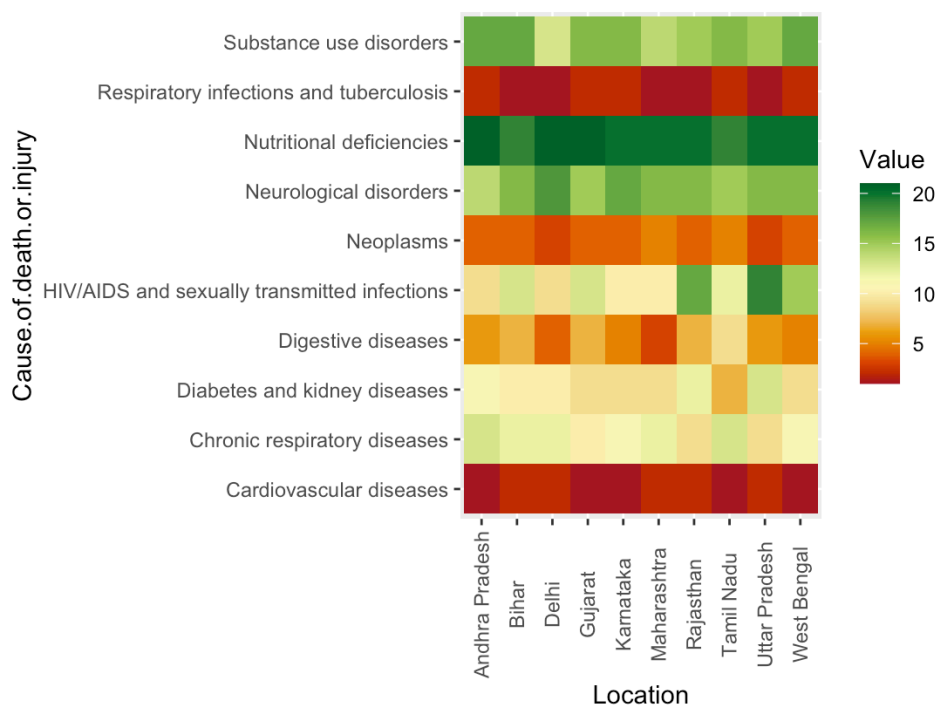
Key Finding and Observations

1. Population Growth Trend



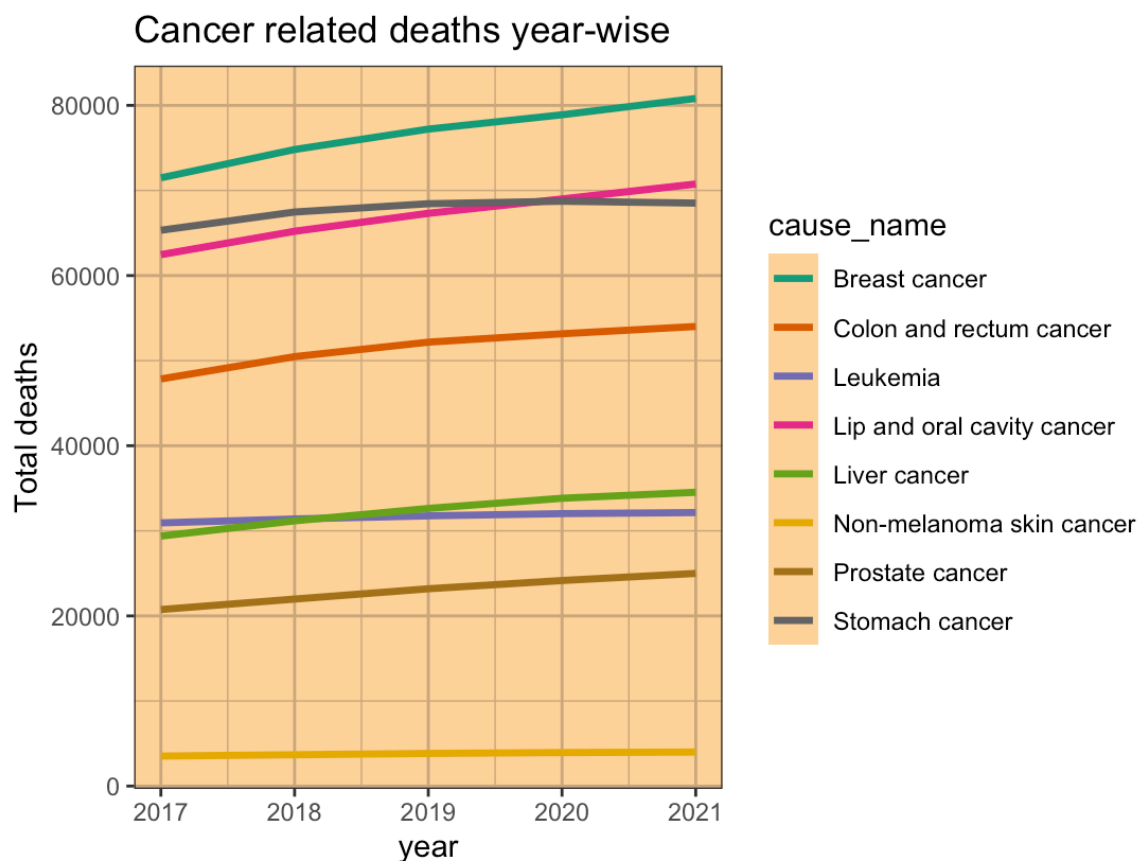
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- The graph illustrates the projected world population over a century. The y-axis represents population in billions, ranging from 0.0 to 1.5 with major grid lines every 0.5 units. The x-axis represents the year, from 2000 to 2100 in 10-year increments. The population curve starts at 1.0 billion in 2000, increases steadily to a peak of approximately 1.7 billion around 2060, and then gradually declines to about 1.5 billion by 2100. A vertical dashed red line is positioned at the year 2060, corresponding to the peak of the population curve.
- | Year | Population (in billions) |
|------|--------------------------|
| 2000 | 1.0 |
| 2010 | 1.2 |
| 2020 | 1.4 |
| 2030 | 1.55 |
| 2040 | 1.65 |
| 2050 | 1.7 |
| 2060 | 1.7 |
| 2070 | 1.65 |
| 2080 | 1.6 |
| 2090 | 1.55 |
| 2100 | 1.5 |

- ## 2. Heat map of causes of death in most populated states



- Cardiovascular diseases, and respiratory diseases are the leading causes of deaths in each state, followed by neoplasms(cancer) and digestive diseases
- Very few Indians die of nutritional deficiencies.
- Delhi has the highest substance-abuse related deaths, Uttar Pradesh has highest respiratory deaths, West Bengal has highest Cardiovascular deaths, while Karnataka has highest Neurological related deaths

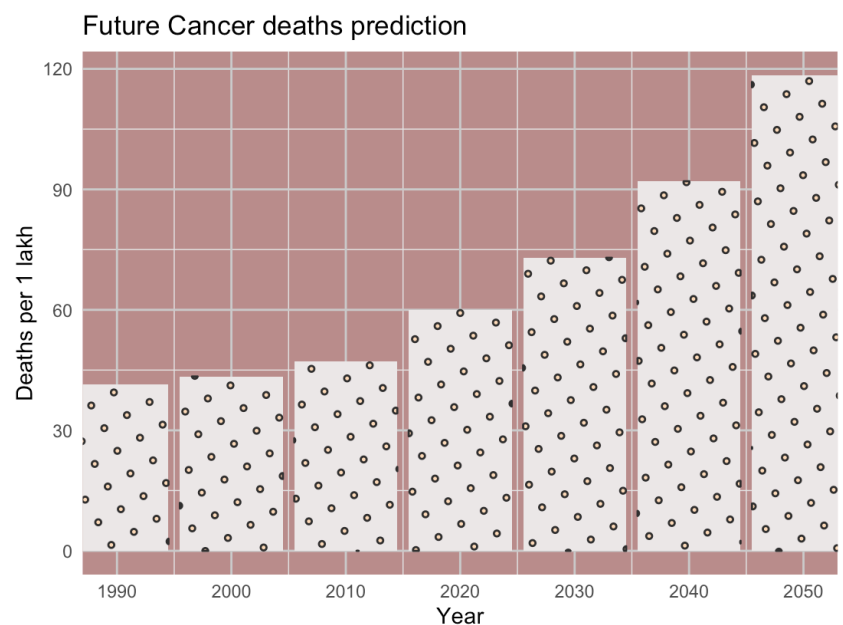
3. Deaths due to different types of cancers



- Breast cancer is the deadliest cancer, followed by oral cancer and stomach cancer.
- Though, deaths due to cancer is increasing every year, skin cancer is not so common in India

4. Cancer Prediction Trend

- For every 1 lakh people, deaths due to cancer went from 40 in 1990 to 62 in 2020. Almost a 50% increase.
- There is steady increase in deaths, reaching to 120 deaths per 1 lakh in 2050,



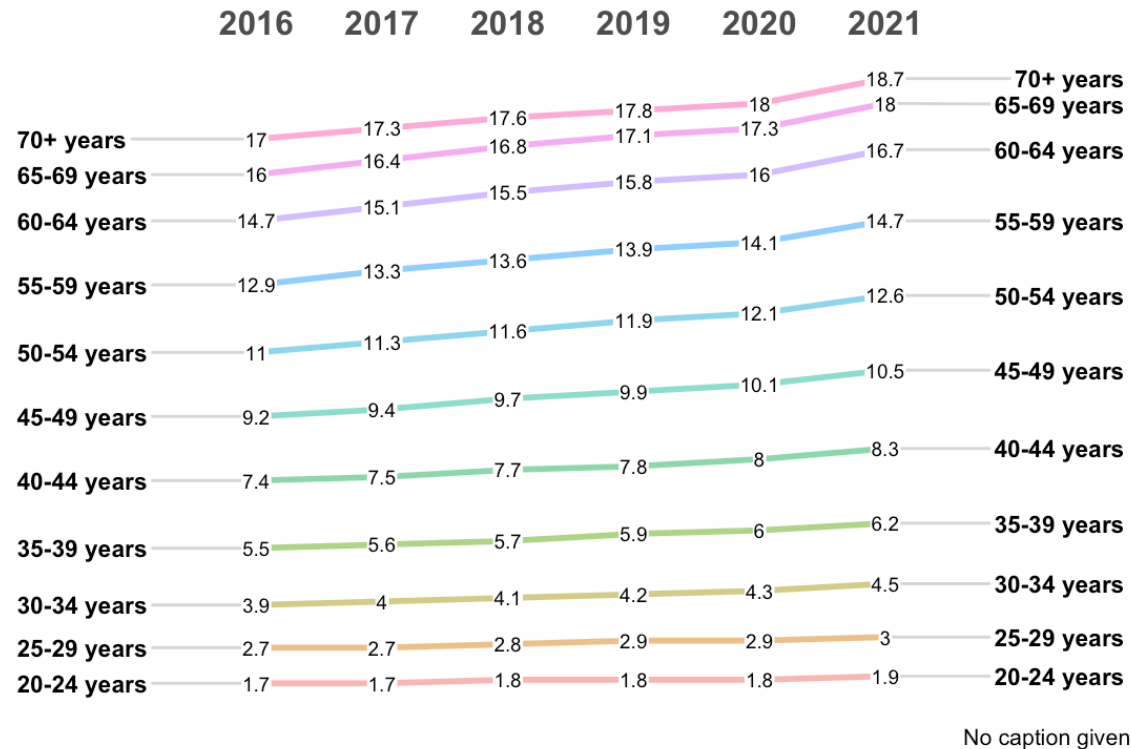
almost a 50% increase in comparison to 2020.

- There seems to be 50% increase in mortality rate every 30 years.

5. Diabetes prevalence rates age-wise

Diabetes rates in India age-wise

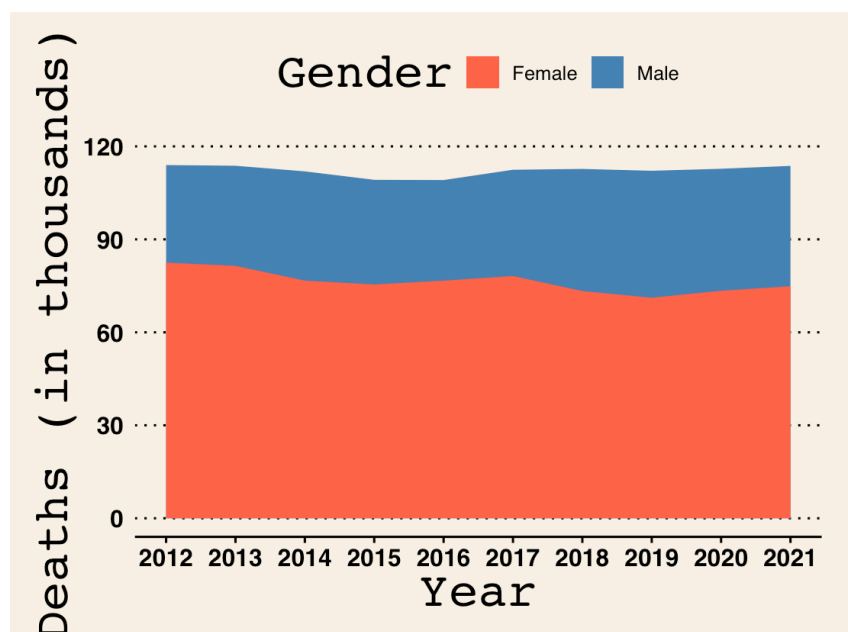
2016-2021



- Risk of diabetes grows with age, with an average change of 2.5% every 10 years
- Difference in diabetes rates are highest between 50s and 60s, meaning most people are diagnosed with diabetes in their 50s.

6. Suicide rates gender wise

- Males tend to commit suicide more than females, with an average male to female suicide ratio of 1.6:1
- Suicides rate for both genders are consistent, with minimal change over the years



References

Inspiration and help for the graphs has been taken from <https://r-charts.com/ggplot2/>.

Inspiration for the ui for RShiny dashboard has been taken from the annual Shiny competition, which is conducted by Posit, where various users come together to show their RShiny skills.

Reproducibility Notes

Required R packages

1. shiny
2. ggplot2
3. dplyr
4. shinythemes
5. shinyWidgets
6. readr

Required csv files (must be in the same directory, with same name)

1. Types of Cancer Deaths.csv
2. heat map pf states.csv
3. self harm.csv
4. Cancer Prediction India rate.csv

Setting working directory

1. Place all csv files in the same folder which has RShiny script
2. Run the following code:

```
setwd("path to app directory")  
shiny::runApp("app.R")
```